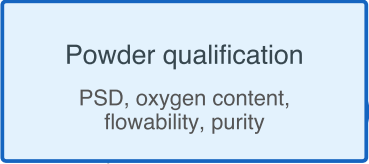
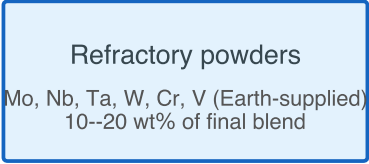


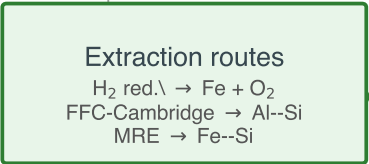
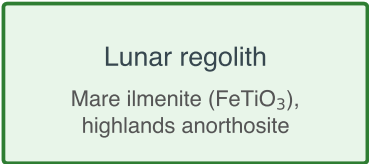
Lunar-metallurgy manufacturing roadmap

Earth-side process calibration (SPARK Phase 2) → ISRU integration → hot-fire validation

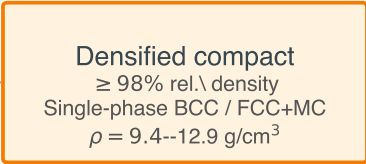
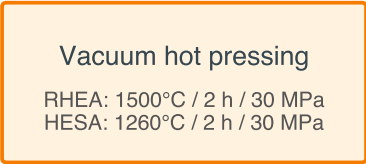
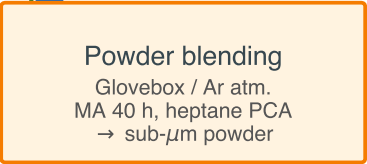
Earth feedstock



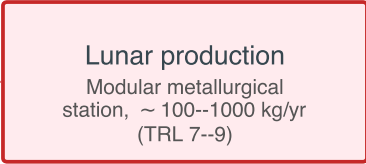
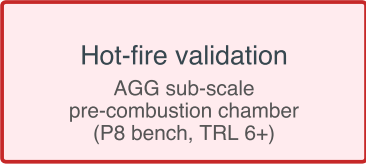
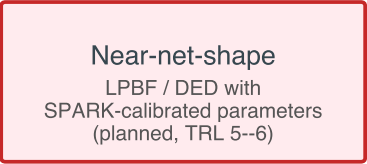
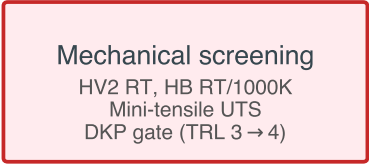
ISRU feedstock



Consolidation + QC



Validation



Stage 1 (TRL 3--5)
Earth-side calibration

Stage 2 (TRL 5--6)
Proxy feedstock + LPBF

Stage 3 (TRL 6+)
Lunar architecture